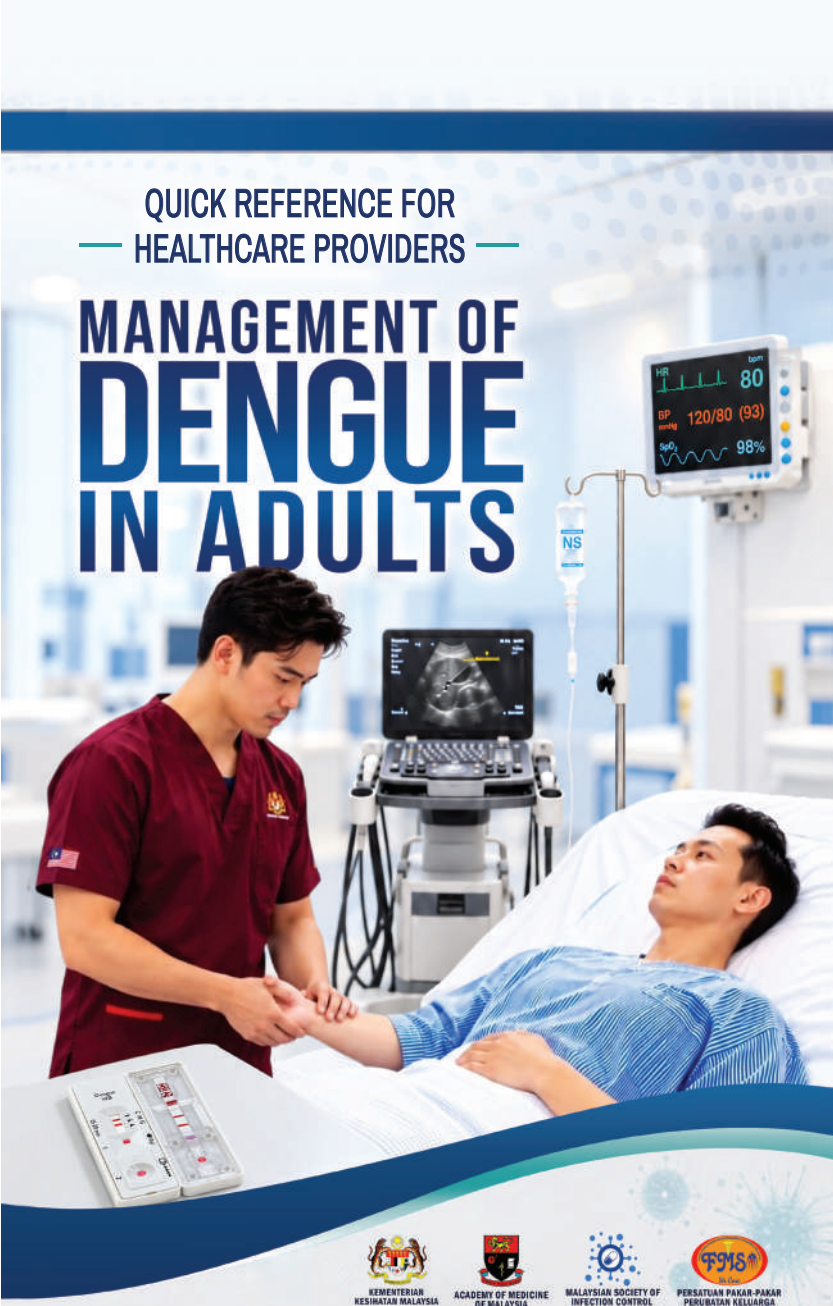
QUICK REFERENCE FOR
HEALTHCARE PROVIDERS

MANAGEMENT OF DENGUE IN ADULTS



KEMENTERIAN KESIHATAN MALAYSIA
ACADEMY OF MEDICINE OF MALAYSIA
MALAYSIAN SOCIETY OF INFECTION CONTROL & INFECTIOUS DISEASES
PERSATUAN PAKAR-PAKAR PERUBATAN KELUARGA MALAYSIA

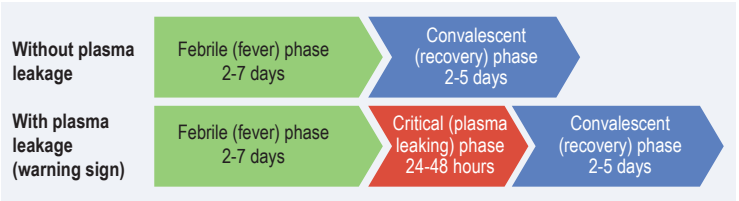
KEY MESSAGES

1. Dengue should be suspected in all patients presenting with acute febrile illness.
2. **Diagnosis**
   - **Dengue Rapid Diagnostic Test (RDT)** with combined NS1 Antigen/IgM/IgG markers is the first-line investigation for diagnosis of dengue.
3. **Outpatient Care for suspected and confirmed dengue**
   - Prescribe **oral fluid therapy (200 ml/hour)** while patients are awake.
   - Ensure the patient passes urine **4 to 6 times each day**.
   - Advise patients to recognise **warning signs in critical phase**.
   - **Avoid NSAIDs and corticosteroids.**
4. **Emergency and Bedside Monitoring**
   - **Point-of-care ultrasound (PoCUS)** should be considered to detect early plasma leakage and assess haemodynamics in patients at risk of deteriorating.
5. **Inpatient Fluid Management**
   - **Febrile Phase**: Prioritise oral fluids; calculate required IV fluids using the **Holliday-Segar formula**.
   - **Critical Phase (without shock)**: Administer a **total fluid quota over 48 hours**, frequently adjusting rates based on clinical signs rather than using a fixed rate.
   - **Dengue Shock: Crystalloids is the preferred fluid** for resuscitation. Colloids can be considered for persistent shock but should not be used as maintenance fluid. Fluid responsiveness and tolerance should be evaluated with dynamic indices.
6. **Managing Complications (Bleeding)**
   - Suspect **occult bleeding** if the patient remains haemodynamically unstable despite receiving adequate fluid resuscitation.
   - Transfuse **packed red blood cells promptly** for clinically significant (overt) bleeding, without waiting for haematocrit to drop.
   - **Routine prophylactic platelet or plasma transfusions are not recommended** in the absence of active bleeding.
7. **Severe Dengue and ICU Care**
   - Promptly refer severe dengue patients to the Intensive Care Unit (ICU).
   - If shock persists despite fluid resuscitation, **noradrenaline is the preferred vasopressor**.
   - Sodium bicarbonate should not be used routinely in all patients with dengue-associated AKI and metabolic acidosis.
8. **Special Populations**
   - **Pregnant patients** require hospitalisation and multidisciplinary management.
   - **Obese patients** in shock require cautious fluid management calculated using their **adjusted body weight**.
   - Fluid management must be highly individualised and closely monitored for patients with **heart failure**, **end-stage kidney disease, and liver cirrhosis**.
   - Maintain a high index of suspicion for dengue in **acutely unwell elderly patients who present with leucopenia**, even without classical features like fever or rash.
9. Dengue vaccination may be considered in all adults, particularly those with comorbidities who are at increased risk of severe dengue, unless contraindicated.

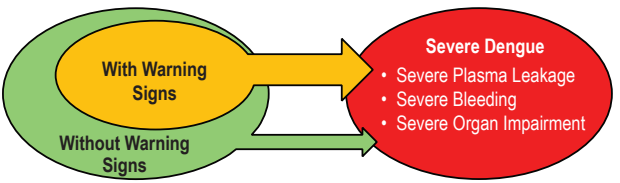
This Quick Reference provides key messages & a summary of the main recommendations in the Clinical Practice Guidelines (CPG) Management of Dengue in Adults (Fourth Edition).
Details of the evidence supporting these recommendations can be found in the above CPG, available on the following websites:
Ministry of Health Malaysia: www.moh.gov.my
Academy of Medicine Malaysia: www.acadmed.org.my
MaHTAS: https://mymahtas.moh.gov.my

CLINICAL PRACTICE GUIDELINES SECRETARIAT
Malaysian Health Technology Assessment Section (MaHTAS)
Medical Development Division, Ministry of Health Malaysia
Level 4, Block E1, Precinct 1,
Federal Government Administrative Centre 62590
Putrajaya, Malaysia
Tel: 603-88831229
E-mail: htamalaysia@moh.gov.my

CLINICAL COURSE OF DENGUE INFECTION



CLASSIFICATION OF DENGUE SEVERITY



Dengue without warning signs (DNWS)	Dengue with warning signs (DWWS)	Severe Dengue (SD)
Person who lives or has travelled to areas with dengue transmission in the last 14 days and presents with fever, usually of 2 to 7 days duration, and at least 2 of the following criteria: • Nausea / vomiting • Rash • Headache/ retro-orbital pain • Myalgia / arthralgia • Petechiae • Leukopenia	Dengue case near or in **critical phase**, that presents with one or more of the following signs: • Lack of clinical improvement or worsening symptoms during the period when fever subsides • Intense abdominal pain or tenderness • Persistent vomiting (>three times per day) • Lethargy, restlessness, irritability or confusion • Fluid accumulation • Postural hypotension • Liver enlargement >2 cm • Reduced or absent urine output for 4–6 hours • Bleeding manifestation • Laboratory findings: ○ Rising haematocrit with rapidly falling platelet count	Dengue case that has one or more of the following manifestations: • Severe plasma leakage leading to shock or respiratory distress • Severe bleeding which is significant enough to require medical intervention • Severe organ involvement (liver impairment, myocarditis, etc.)
	Requires strict monitoring and immediate medical intervention	

INTEGRATION OF MTS WITH DENGUE CLASSIFICATION AND RECOMMENDED MANAGEMENT

MTS Level	Zone	Dengue Classification	PAHO 2020 Recommended Management*
Level 1 Resuscitation	Red	Decompensated dengue shock syndrome	Immediate resuscitation, ICU admission (Group C)
Level 2 Emergency	Red	Compensated dengue shock syndrome	Emergency treatment, IV fluid therapy and admission (Group C)
Level 3 Urgent	Yellow	Dengue with warning signs or without warning sign but unstable co-morbidity	IV fluid therapy, admission, management of associated conditions (Group B2)
Level 4 Early care	Green	Dengue without warning signs with stable comorbidity	Observation or admission, management of associated conditions (Group B1)
Level 5 Routine care	Outpatient	Dengue without warning signs, comorbidity and social risk	At-home outpatient management with follow-up (Group A)

INTERPRETATION OF RAPID DIAGNOSTIC TESTS FOR DENGUE

Clinical History	Test	Result			Interpretation
		NS1	IgM	IgG	
History of fever less than 5 days	Rapid Diagnostic Test (RDT) for Dengue	⊕	⊖	⊖	Acute dengue infection
		⊕	⊕	⊖	Acute dengue infection
		⊕	⊖	⊕	Acute dengue infection or secondary infection
		⊖	⊖	⊖	Not dengue, but if history suggestive of dengue, repeat RDT
History of fever more than 5 days	RDT for Dengue	⊕	⊕	⊖	Acute dengue infection. High risk for severe dengue
		⊖	⊕	⊖	Acute or recent dengue infection
		⊖	⊕	⊕	Acute, recent or secondary dengue
		⊖	⊖	⊕	Unable to rule out secondary dengue, proceed with dengue IgG high-titre
		⊖	⊖	⊖	Not dengue infection

DDENGUE MANAGEMENT BASED ON SEVERITY

Category	Group A	Group B1	Group B2	Group C
Severity classification	**Dengue without warning signs (DNWS)**	**Dengue without warning signs (DNWS)**	**Dengue with warning signs (DWWS)**	**Severe dengue (SD)**
Group criteria	• Stable general well-being (able to drink adequately, good urine output and able to ambulate) • Absence of warning signs in critical phase • Haemodynamically stable • Stable blood investigation • Absence of unstable comorbidities or social factors	Presence of **comorbidities**: * • Hypertension • Diabetes mellitus • Asthma/ COPD • Chronic kidney disease • Haemolytic diseases • Chronic liver disease • Peptic ulcer disease or gastritis of any etiology • Being treated with anticoagulants/ antiplatelet • others **High risk group**: • Pregnancy • ≥65 years old • Obesity • Patients with special needs **Social risk**: • The patient lives alone or far from health facilities • Does not have transportation *Unstable comorbidities must be referred for assessment	Dengue case near or in **critical phase**, that presents with one or more of the following signs: 1. Lack of clinical improvement or worsening of symptoms when fever subsides 2. Intense abdominal pain or tenderness 3. Persistent vomiting (>three times per day) 4. Lethargy, restlessness, irritability or confusion 5. Fluid accumulation 6. Postural hypotension (lipothymia) 7. Liver enlargement >2 cm 8. Progressive increase in haematocrit with drop in platelets 9. Bleeding manifestation 10. Reduced or absence of urine output 4-6 hours	Every dengue case that has one or more of the following manifestations: • Severe plasma leakage leading to shock or respiratory distress • Severe bleeding which is significant enough to require medical intervention • Severe organ involvement (liver impairment, myocarditis, encephalitis etc.)
Management of care	**Outpatient monitoring**	**Possible referral to hospital** May require close observation and treatment of their associated condition.	**Admit to hospital**	**Intensive Care Unit** Requires emergency treatment

The decision for referral and admission must not be based on a single clinical parameter instead should depend on the total assessment of the patient.

DENGUE CLINICAL ASSESSMENT IN PRIMARY CARE

Steps	Key Components
History Taking	• Day of illness and date of fever onset. • Symptoms suggestive of dengue and living in/travelling to a dengue endemic area • Presence of dengue warning signs • Daily function such as daily activity, oral fluid intake and urine output. • Comorbidities • Special group populations such as pregnancy, elderly > 65 years old • Social factors affecting follow-up and access to care • Dengue vaccination status and date of vaccination.
Clinical Examination	• General condition, mental status and Glasgow Coma Scale, hydration status • Haemodynamic status such as blood pressure, pulse pressure, pulse rate, respiratory rate and temperature. • Anthropometry such as weight, height and BMI • Assessment of CCTVR (Colour, Capillary Refill Time (CRT), Temperature, Pulse Volume, Pulse Rate). • Assessment of shock. • Respiratory status (tachypnoea, pleural effusion) • Abdominal examination (tenderness, hepatomegaly, ascites) • Assess bleeding tendency (skin, mucosa, injection sites)
Investigations	• Point-of-care test for dengue (Dengue Rapid Diagnostic Test). If unable to perform, patients should be referred to nearest facility where it can be done. • Full Blood Count (FBC) • AST (if available) • Ultrasound (PoCUS - if trained and available)
Diagnosis	• **Suspected dengue**: Fever + ≥2 symptoms +/- supportive lab findings • **Confirmed dengue**: laboratory evidence of dengue infection • A complete diagnosis should include the following: ○ Day of dengue fever ○ Dengue phase: febrile, critical or recovery ○ Classification of dengue ○ Presence of shock ○ Comorbidities or special populations
Management Decision, Patient Education & Safety Netting	• Notify the district health office • Determine suitability for outpatient management (**Group A**), referral (**Group B1**) or admission (**Group B2** and **C**). Refer to **Table on Dengue Severity and Management**. • If managed at an outpatient department (**Group A - DNWS**) ○ Arrange daily follow-up and monitoring until patient is afebrile for **24 - 48 hours without antipyretics** ○ Provide **Outpatient Dengue Monitoring Record** (**Appendix 5**) ○ Educate the patient on DOs and DON'Ts, warning signs and fluid intake. ○ Advise immediate return if the patient has warning signs.

FLUID QUOTA FOR DENGUE IN CRITICAL PHASE WITHOUT SHOCK

Fluid Quota

For the entire critical phase, the recommended total fluid volume is **Maintenance + 5% deficit**, **where a 5% deficit is equivalent to 50ml/kg**. This fluid quota includes both oral and intravenous fluid intake.

Important Considerations:

- Not all patients will require the full fluid quota Maintenance + 5% deficit; many may need less as the rate, peak, and duration of plasma leakage vary between individuals.
- 5% deficit represents the expected plasma volume loss during critical phase
- Crystalloid is the intravenous fluid of choice for maintenance therapy.
- This entire amount is to be administered over 48 hours.

PREDICTORS OF SEVERE DENGUE

- Patient's factor:
  - Older age, Overweight/obesity (BMI >25 kg/m²), Co-morbid conditions: Type-2 DM with poor glycaemic control, presence of one or more co-morbidities
- Clinical Signs:
  - Presence of multiple warning signs,
  - Gastrointestinal bleeding
  - Clinical features suggestive of plasma leakage
  - Altered mental status
  - In-hospital complications e.g., acute kidney injury or pneumonia
- Laboratory Markers:
  - Hematologic: Early decline in platelet count in febrile phase, severe thrombocytopenia (platelet count <50 × 10^9/L), leucocytosis or rising haematocrit with rapid drop in platelet
  - Biochemical: Early elevation of AST/ALT in febrile phase or severe hepatitis >1000 U/L
  - Coagulation: Prolonged PT
- Imaging: Gallbladder wall oedema/ increased gallbladder wall thickness, pleural effusion, ascites and/or organomegaly

Severe dengue is predicted by a **combination** of the above factors.

FEATURES COMPARING COMPENSATED AND DECOMPENSATED SHOCK

Normal Circulation	Compensated Shock	Decompensated/ Hypotensive Shock
Clear consciousness	Clear consciousness	Lethargy, restlessness, irritability or confusion
CRT≤3sec	CRT>3 sec	CRT>3 sec
Warm peripheries	Cool peripheries	Cool peripheries
Heart rate (HR) normal	Tachycardia	Severe tachycardia or bradycardia in late shock
Blood pressure (BP) normal	Normal systolic pressure with raised diastolic pressure	Hypotension
Normal pulse pressure	Narrowing pulse pressure (20 - 30 mmHg)	Narrowed pulse pressure (<20 mmHg)
Normal respiratory rate	Tachypnoea	Tachypnoea or Kussmaul's breathing
Normal urine output	Reduced urine output	Oliguria or anuria

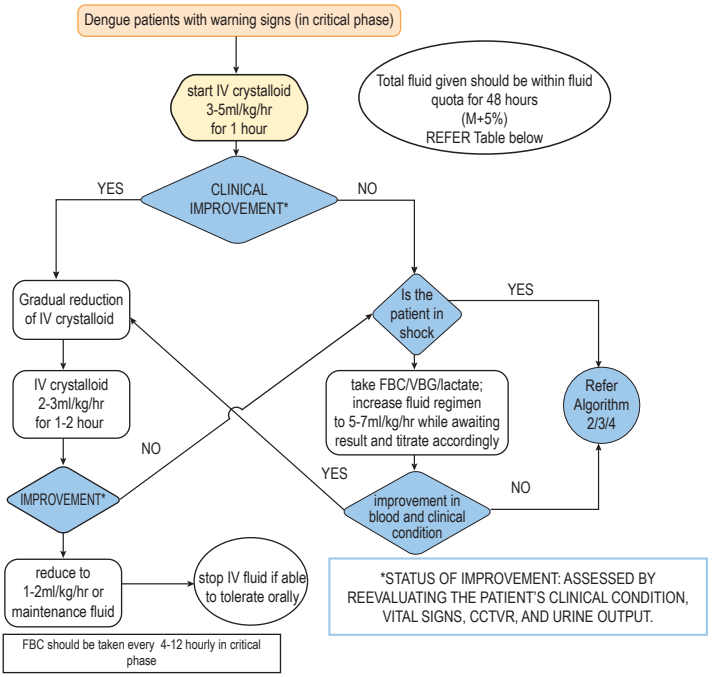
INDICATORS OF EFFECTIVE FLUID RESUSCITATION

All these parameters should be interpreted collectively and not in isolation

A. Clinical Parameters
- Improvement of general well-being/mental state
- Warm peripheries with stable BP
- Capillary refill time ≤ 3sec
- Improving pulse pressure
- Reducing tachycardia
- Improving urine output
- Less tachypnoea

B. Laboratory Parameters
- Appropriate decrease in HCT
- Improvement in metabolic acidosis and lactate

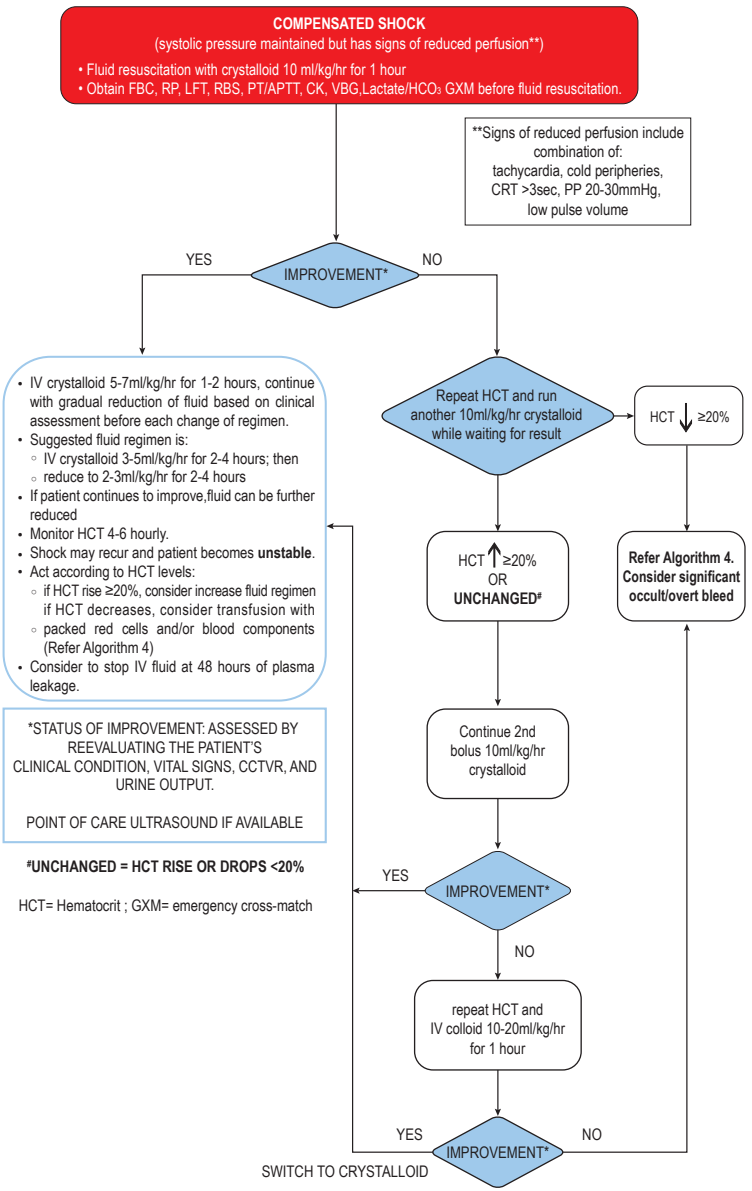
ALGORITHM 1: MANAGEMENT OF DENGUE PATIENTS WITH WARNING SIGNS (GROUP B2)



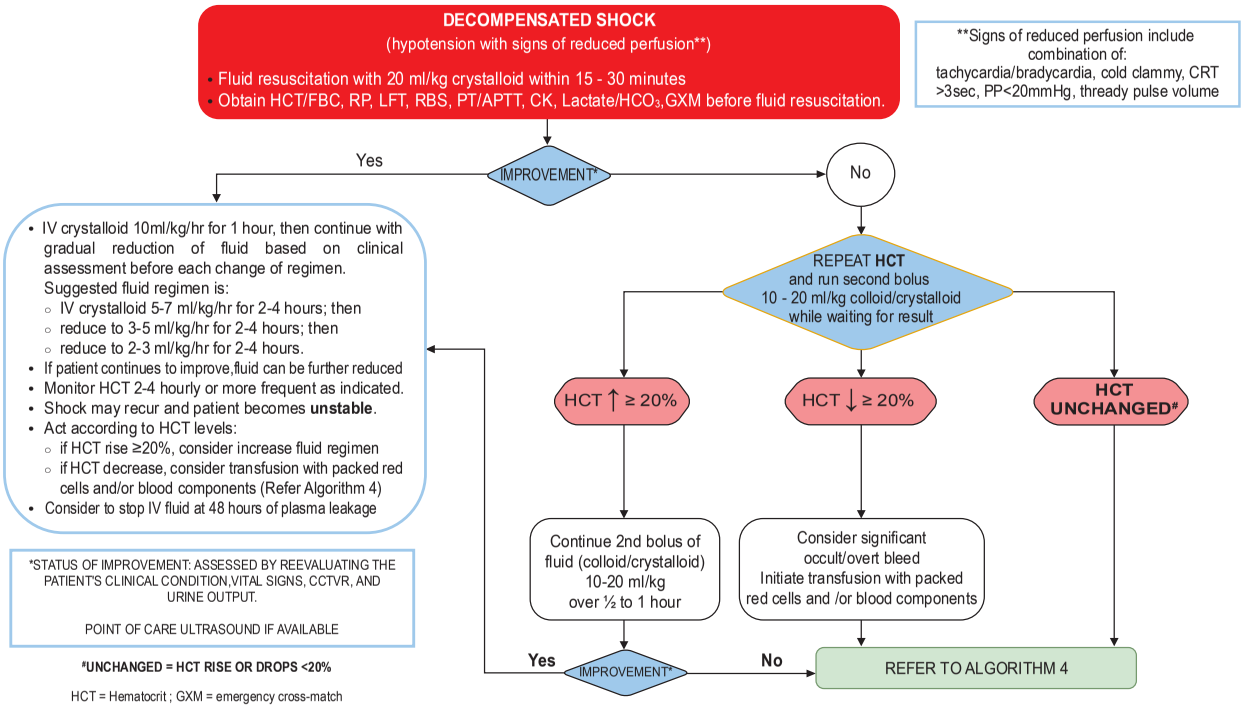
Body weight (kg)*	Fluid quota M+5% (Oral and IV) over 48 hours (ml)	Rate of IV fluid regime (ml/hr)				
		R1 (1ml/kg/h)	R2 (2ml/kg/h)	R3 (3ml/kg/h)	R5 (5ml/kg/h)	R7 (7ml/kg/h)
30	3200	30	60	90	150	210
35	3600	35	70	105	175	245
40	3900	40	80	120	200	280
45	4300	45	90	135	225	315
50	4700	50	100	150	250	350
55	5000	55	110	165	275	385
60	5400	60	120	180	300	420
65	5700	65	130	195	325	455
≥ 70	6000	70	140	210	350	490

*weight limit at 70kg

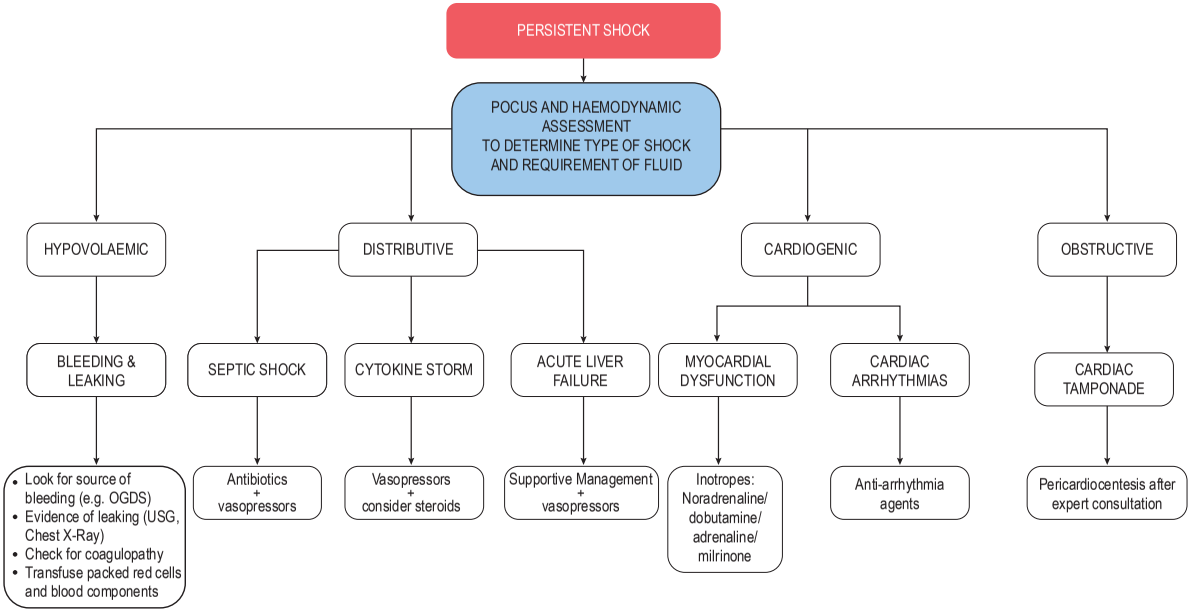
ALGORITHM 2: FLUID MANAGEMENT IN COMPENSATED SHOCK (GROUP C)



ALGORITHM 3: FLUID MANAGEMENT IN DECOMPENSATED SHOCK (GROUP C)



ALGORITHM 4: MANAGEMENT OF REFRACTORY SHOCK (GROUP C)

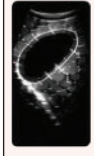
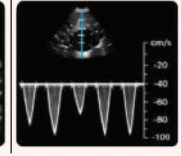
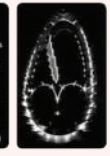
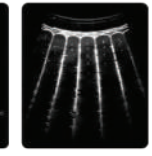


Patient may have more than one type of shock

OGDS= oesophagogastroduodenoscopy; USG= ultrasonography; POCUS = point of care ultrasound

POCUS GUIDED DENGUE MANAGEMENT

A PHASE GUIDED APPROACH

	FEBRILE PHASE (Day 2 - 7)	CRITICAL PHASE (Day 3 - 7)				RECOVERY PHASE (Day 7 onwards)	
1 PHASE							
2 POTENTIAL COMPLICATIONS	DEHYDRATION	LEAKING	BLEEDING	SHOCK	ORGAN DYSFUNCTION	REABSORPTION	FLUID OVERLOAD
3 POCUS FINDINGS	SMALL / FLAT IVC (<1.5 cm)	GBWT >3 mm	PLEURAL EFFUSION	ASCITES	LOW LVOT VTI	REDUCED CARDIAC FUNCTION	DILATED IVC (POOR VARIABILITY)	B-LINES (FLUID OVERLOAD)
								
	HYPERDYNAMIC HEART WITH SMALL LV, RV							
								
4 POCUS GUIDED MANAGEMENT	• Assess volume • Guide fluids • Screen for early leakage	• Detect leakage • Assess hemodynamics • Differentiate type of shock • Guide fluids / blood / vasoactives • Serial reassessment				• Confirm resolution • Detect fluid overload • De-escalate fluids	

IMPORTANT: SIGNS CAN COEXIST AND OVERLAP ACROSS DIFFERENT PHASES, CLINICAL JUDGEMENT AND SERIAL POCUS ARE ESSENTIAL.

POCUS USE:
- ✓ When clinically indicated
- ✓ By trained operators
- ✓ Where resources are available

IMPORTANT:
POCUS COMPLEMENTS CLINICAL ASSESSMENT. ALWAYS INTEGRATE WITH CLINICAL JUDGMENT AND PATIENT CONTEXT

POCUS CAN HELP TO:
- ✓ Detect hypovolemia early
- ✓ Identify plasma leakage and complications
- ✓ Guide fluid therapy and monitor response
- ✓ Support safe de-escalation during recovery

Note: PoCUS - Point-of-care Ultrasound, IVC - Inferior Vena Cava, LVOT VTI - Left Ventricular Outflow Tract Velocity Time Integral, GBWT - Gallbladder Wall Thickening, B-lines - Lung comet-tail artifacts.

REFERENCES:
1. World Health Organization. Dengue Guidlines For Diagnosis, Treatment, Prevention And Control. Geneva: WHO; 2009.
2. World Health Organization. WHO guidlines for clinical management of arboviral diseases: dengue, chikungunya and yellow fever. Geneva: WHO; 2025.